

AFRICA southwest

Hoba West

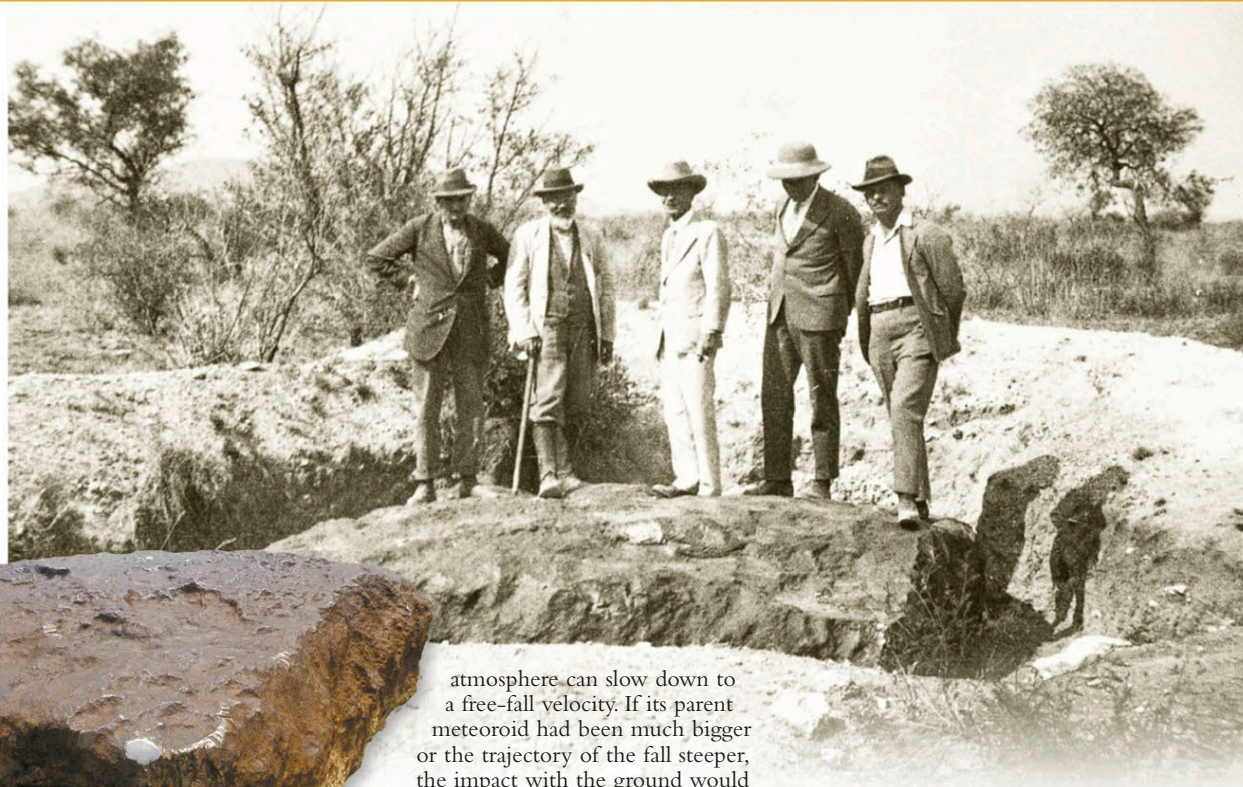


LOCATION Grootfontein, Namibia
TYPE Iron
MASS 66 tons
DATE OF DISCOVERY 1920

The largest meteorite to have been found on Earth, Hoba West measures 8.9 x 8.9 x 3 ft (2.7 x 2.7 x 0.9 m). It consists of 84 percent iron and 16 percent nickel. Hoba West has never been moved from where it landed. In the past, enterprising individuals tried to recover this valuable lump of “scrap” metal. To protect it from damage and sample-taking, the Namibian Government has declared it to be a national monument. Hoba West represents the maximum mass that the Earth’s

RUSTING AWAY

The Hoba West meteorite weighed about 66 tons when it was discovered, but it has started to rust away and today weighs less than 60 tons.



atmosphere can slow down to a free-fall velocity. If its parent meteoroid had been much bigger or the trajectory of the fall steeper, the impact with the ground would have been much faster. This would have led to the destruction of most of the meteorite and the production of a crater in the Earth’s surface. Large lumps of surface iron, such as Hoba West, are hard to overlook.

LARGEST KNOWN METEORITE

A team of scientists from Kings College, London, UK, poses on top of Hoba West in the 1920s. Standing second from the left is Dr. L. J. Spencer, who became Keeper of Minerals at the British Museum, London, in the 1930s.

AFRICA south

Cold Bokkeveld



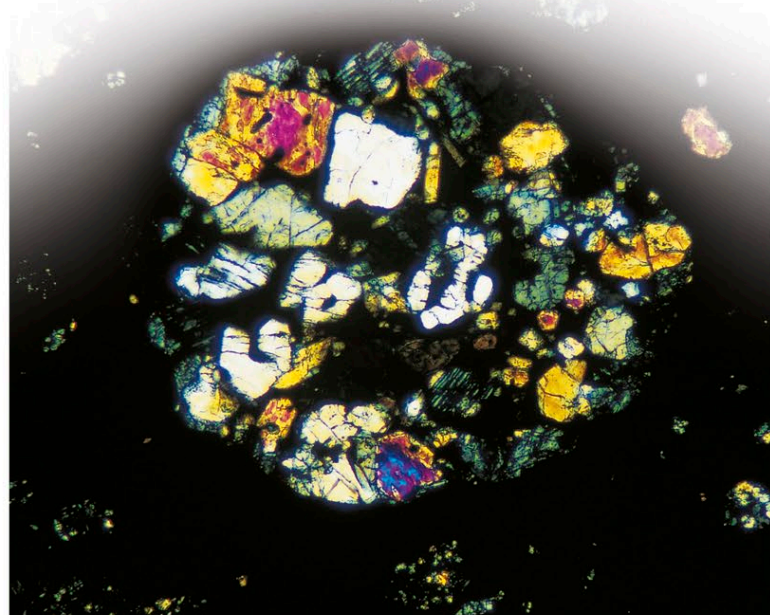
LOCATION Western Cape, South Africa
TYPE Stony
MASS About 8.8 lb (4 kg)
DATE OF DISCOVERY 1838

This meteorite is a perfect example of a stony chondrite, a class of primitive meteorite that makes up almost 90 percent of those found so far. They consist of silicate, metallic, and sulfide minerals and are thought to represent the material from which the Earth was formed. They contain tiny, spherical

chondrules cemented into a rocky matrix. These rocky droplets solidified extremely quickly from a starting temperature of at least 2,600°F (1,400°C). Chondrules contain a mixture of imperfect crystals and glass. Cold Bokkeveld is carbonaceous, which means that it contains compounds of carbon, hydrogen, oxygen, and nitrogen. These are the main constituents of living cells. Carbonaceous chondrites thus contain the building blocks of life.

WATER FROM STONE

This tiny chondrule is surrounded by a water-rich matrix (shown in black). Cold Bokkeveld contains about 10 percent water by mass, which would be released if it was heated.



AUSTRALIA west

Mundrabilla



LOCATION Nullarbor Plain, Western Australia
TYPE Iron
MASS About 18 tons
DATE OF DISCOVERY 1911

Mundrabilla is on the Trans-Australian railroad line in a featureless desert. Three small irons were found there in 1911 and 1918. Renewed interest in 1966 led to the discovery of two meteorites weighing 5 and 11 tons. Mundrabilla took many millions of years to solidify, and

it offers a rare chance to investigate the formation of alloys at low gravity. A 100-lb (45-kg) core of one of the meteorites (below) is undergoing computer X-ray analysis by NASA.



UNDER INVESTIGATION

ANTARCTICA

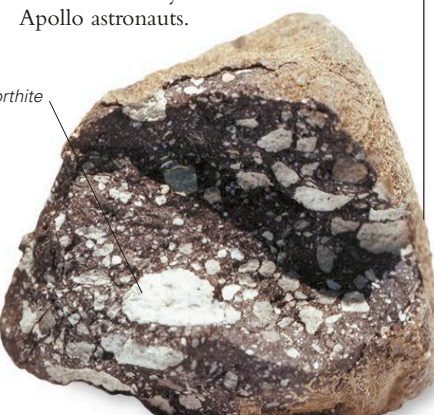
ALH 81005



LOCATION Allan Hills, Antarctica
TYPE Stony
MASS 1.1 oz (31.4 g)
DATE OF DISCOVERY 1982

ALH 81005 is a lunar meteorite. About 36 have been discovered, a mere 0.08 percent of the present total. The cosmic-ray damage they have suffered indicates that they have been blasted from the surface of the

Moon by a meteorite impact in the last 20 million years. The main mineral is anorthite (calcium aluminum silicate), which is very rare in asteroids. The composition of these stony meteorites is very similar to that of the lunar-highland rocks brought back to Earth by the Apollo astronauts.



MOON ROCK

This golf-ball-sized rock was found by the US Antarctic Search for Meteorites program in 1982. It was the first meteorite to be recognized as being of lunar origin.